



P@SHA ICT AWARDS 2026 · STUDENT INNOVATION CATEGORY

ConceptIntel

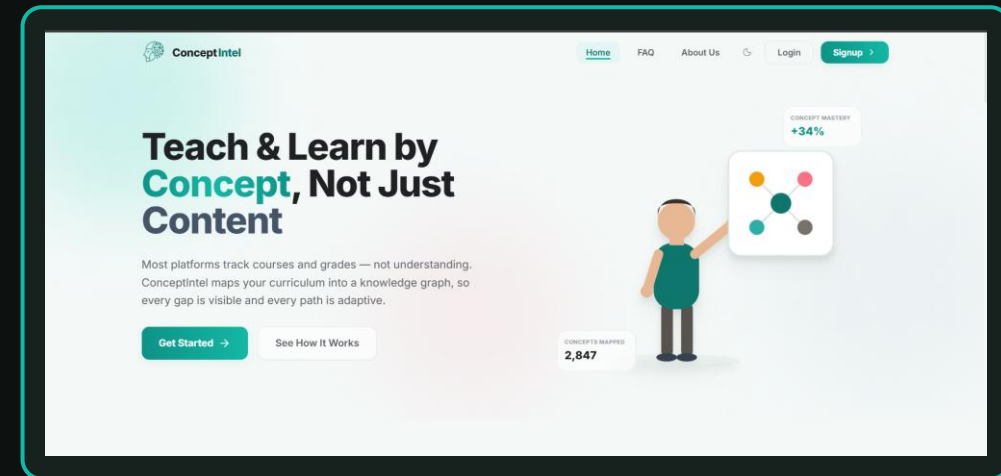
A Knowledge Graph–Based Concept Intelligence Platform for Adaptive Higher Education

Real knowledge graphs. Real AI extraction. Five real user roles — running today, not mocked up.

- **Mamoona Latif** 231662 — Auth, Content Upload, AI Content Generation, Notifications
- **Laiba** 231627 — Course Mgmt, Knowledge Graph, Analytics, Gamification
- **Muhammad Kashif** 231568 — Enrollment, Supervision, Assignment Evaluation, Adaptive Engine

Supervisor: **Dr. Sumaira Farid**

Air University, Islamabad | Faculty of Computing & AI



Live at ConceptIntel — real product, not a mockup

Rethinking Digital Learning

- Universities run on LMS platforms that manage courses and record grades — but never measure understanding.
- Knowledge is hierarchical: you cannot grasp acceleration without first understanding velocity and time.
- ConceptIntel maps every course into a knowledge graph, so gaps are visible before a final score ever hides them.

“Today's systems measure performance, but they do not measure understanding.”

3

Pilot Subjects

5

User Roles

12

Planned Modules

1

Unified Platform

THE PROBLEM

The Challenge in Higher Education

- LMS platforms manage courses and grades — but never measure conceptual understanding.
- Students advance without mastering prerequisites, so gaps compound silently over semesters.
- Teachers see overall scores — never which specific concepts a student is actually struggling with.
- Generic AI chatbots explain topics, but stay disconnected from the actual course curriculum.

Student → LMS → Grades



No concept-level visibility for teacher or student.

**Student → AI Chatbot
→ Generic Answer**



Not tied to the course, curriculum, or prior mastery.

Existing Solutions vs. ConceptIntel

	Course Mgmt	AI Assist	Know. Graph	Adaptive Learning	Teacher Review	Concept Analytics
Google Classroom	✓	—	—	—	✓	—
Moodle	✓	—	—	—	✓	—
Coursera	✓	—	—	~	—	—
Khan Academy	—	—	—	~	—	—
ChatGPT	—	✓	—	—	—	—
ConceptIntel	✓	✓	✓	✓	✓	✓

ConceptIntel is the only platform in this comparison that checks every single box.

Introducing ConceptIntel (1 of 2)



AI-Powered Academic Intelligence

One platform for students and teachers, built around concepts — not just courses.



Knowledge-Graph-Driven Learning

Course material becomes an interconnected map of concepts and prerequisites.



Retrieval-Augmented Generation

Concept extraction is grounded directly in each course's own uploaded material — not guesswork.

Introducing ConceptIntel (2 of 2)



Adaptive Learning Engine

Personalized paths that reinforce exactly the concepts each student is weak on.



Teacher-in-the-Loop

Every AI-generated concept and knowledge graph is reviewed and approved before students see it.



Curriculum Intelligence

Knowledge graphs are shared and reused across every instructor teaching the same course.

GOALS

Objectives (1 of 2)

- Develop a web-based AI platform supporting both students and teachers.
- Extract key concepts from course materials into a knowledge graph.
- Track concept-level mastery for every student, continuously.
- Generate adaptive learning materials — with explainable feedback.

GOALS

Objectives (2 of 2)

- Give teachers actionable insight into learning gaps and bottlenecks.
- Enable teacher supervision and approval of all AI-generated content.
- Reinforce weak concepts through personalized, gamified exercises.
- Support continuous improvement through feedback-driven regeneration.

Prototype Scope

Three real subjects, chosen to prove prerequisite-based concept relationships actually work end to end:



Applied Physics

Foundational prerequisite subject



Digital Logic Design

Builds directly on Applied Physics



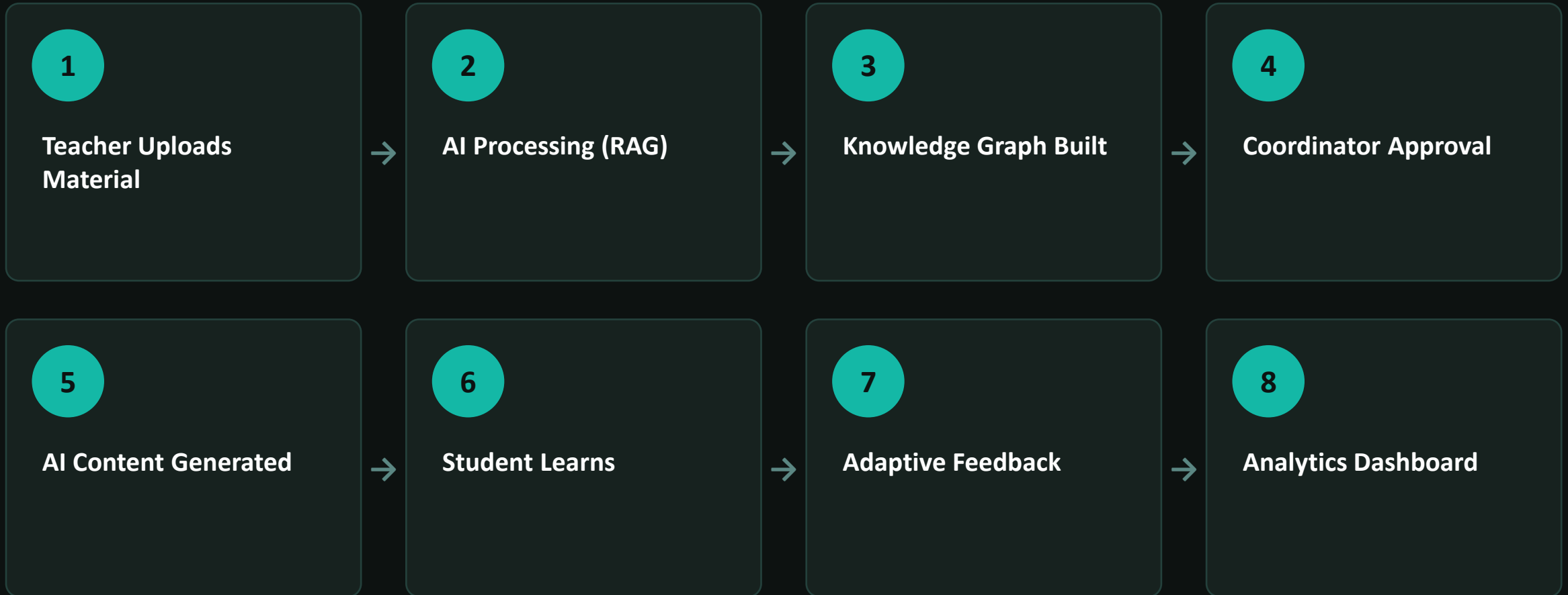
Calculus & Analytical Geometry

Independent third pilot subject

- Teachers upload slides, PDFs, textbooks, CLOs, PLOs, and HEC outlines.
- Graphs are shared across every teacher on the same course — and reused by future instructors.

HOW IT WORKS

End-to-End Workflow



Core Features — Live Today



Knowledge Graph Builder

LIVE



Content Upload + RAG

LIVE



AI Concept Extraction

LIVE



5-Tier Role-Based Access

LIVE



Coordinator Approval

LIVE



AI Assistant Chat

LIVE

Core Features — Live & Upcoming



Class Stream

LIVE



Assignments

LIVE



Notification Center

LIVE



Adaptive Learning Engine

UPCOMING



Analytics Dashboard

UPCOMING



Gamification

UPCOMING

System Modules (Part 1 of 2)

7.1 User Authentication

Sign-up, login, role-based access for 5 roles.

LIVE

7.2 Course Management

Create/edit/delete courses with scheduling & status.

LIVE

7.3 Enrollment

Enrollment codes, join links, prerequisite checks.

LIVE

7.4 Content Upload

AI + RAG pipeline extracts concepts & objectives.

LIVE

7.5 Knowledge Graph

Build, edit, and share concept dependencies.

LIVE

7.6 Content Generation

Generate quizzes and study guides from concepts.

UPCOMING

System Modules (Part 2 of 2)

7.7 Teacher Supervision

Review and approve AI-generated content.

LIVE

7.8 Assignment Evaluation

Concept-level grading with explainable feedback.

IN PROGRESS

7.9 Adaptive Engine

Personalizes each student's learning path.

UPCOMING

7.10 Analytics Dashboard

Concept-mastery heatmaps for teachers.

UPCOMING

7.11 Gamification

Mini-games tied to concept mastery.

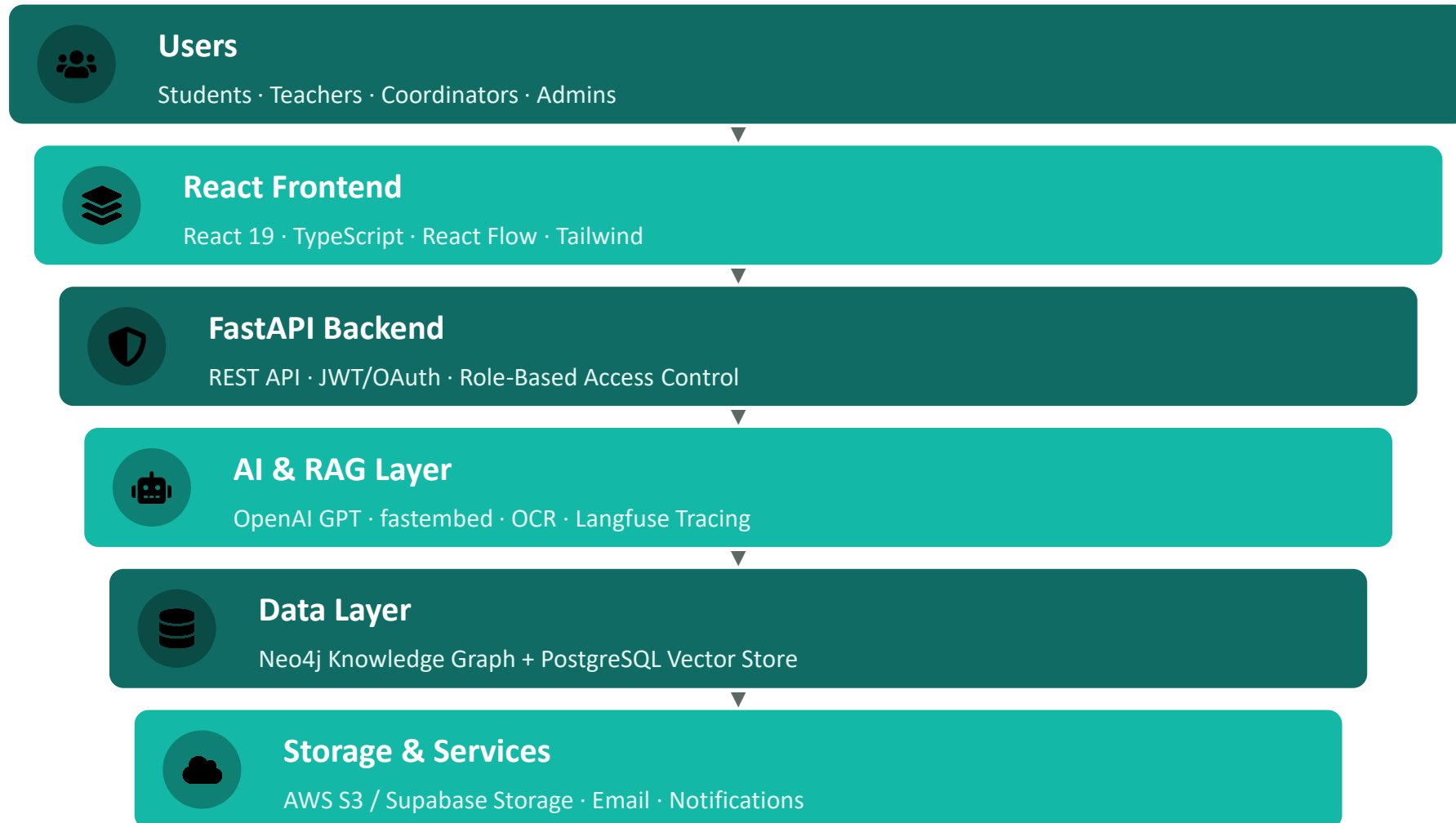
UPCOMING

7.12 Notification Module

Real-time alerts for content and feedback.

LIVE

System Architecture



Technology Stack

Frontend

- React 19
- TypeScript
- TanStack Query
- React Flow
- Tailwind

Backend

- FastAPI
- SQLAlchemy
- PostgreSQL
- Uvicorn

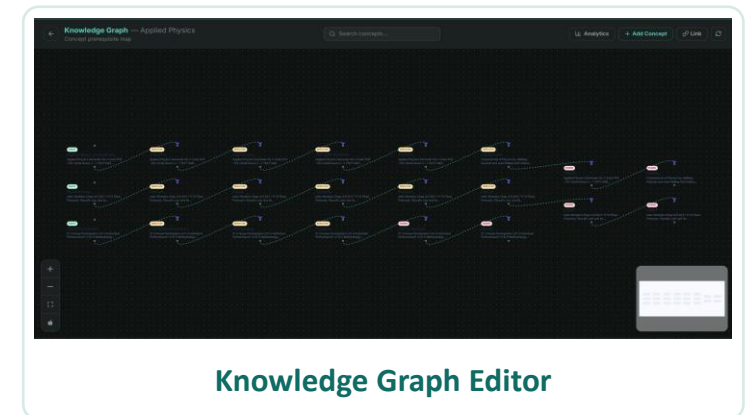
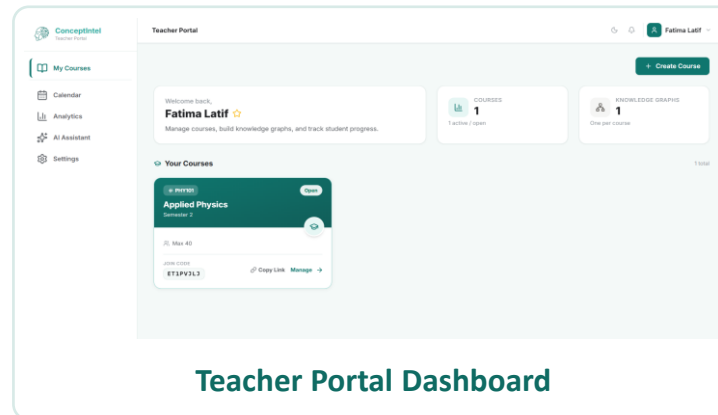
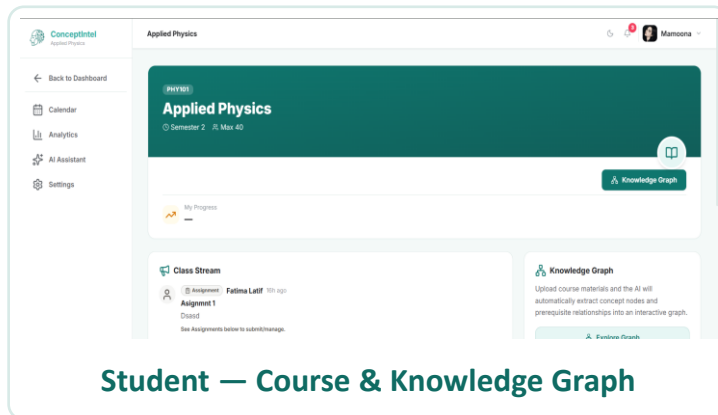
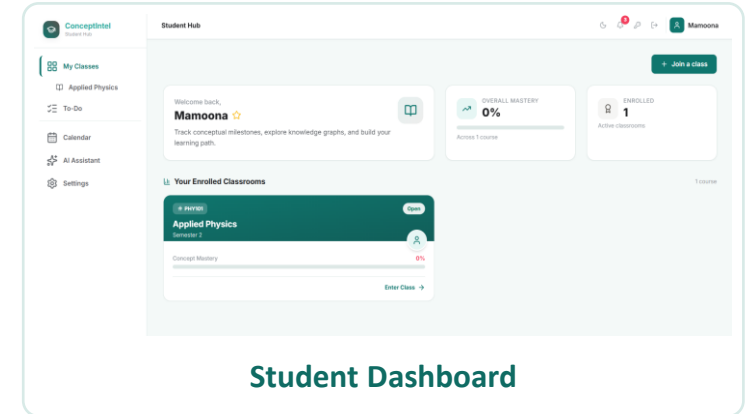
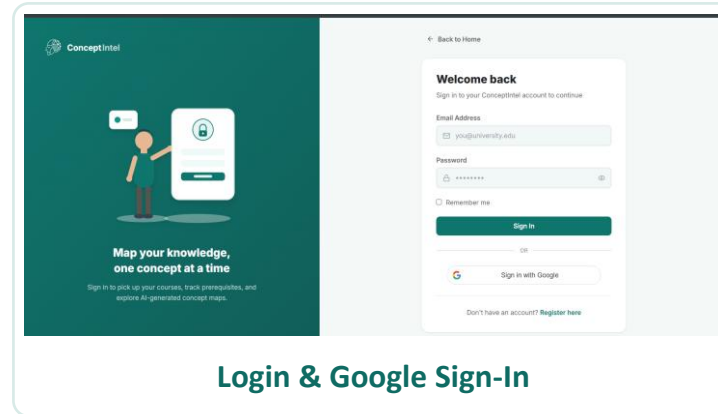
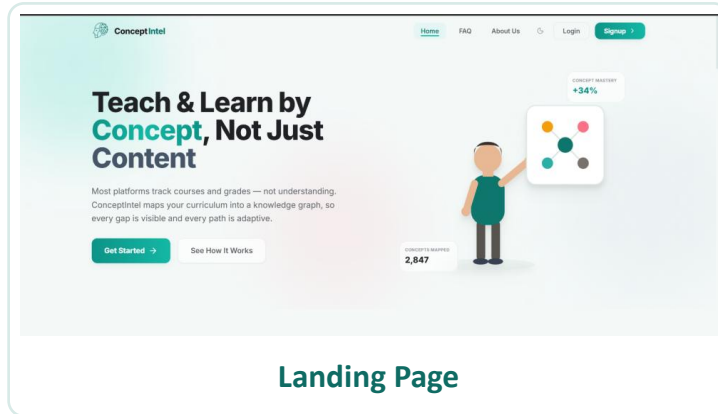
AI / ML

- OpenAI GPT
- fastembed
- Tesseract OCR
- Langfuse

Graph & Infra

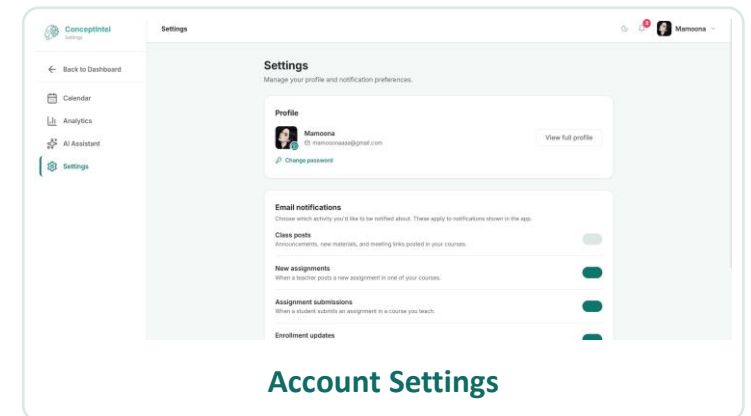
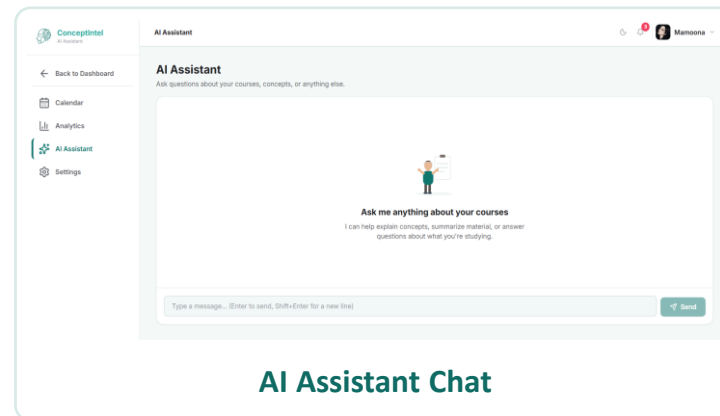
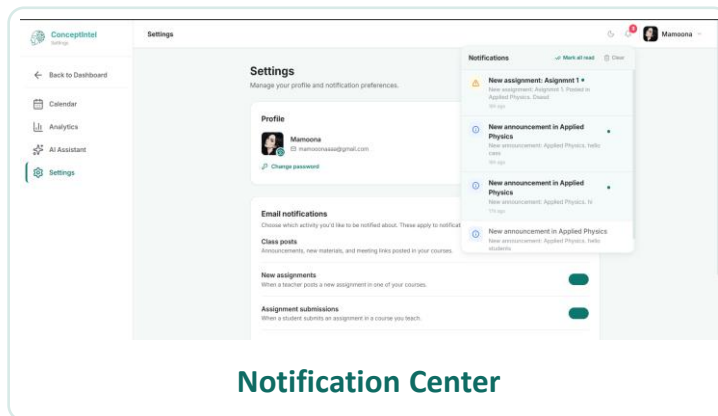
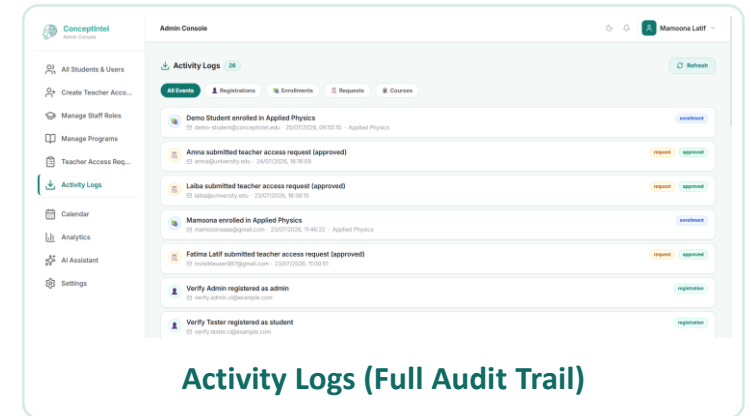
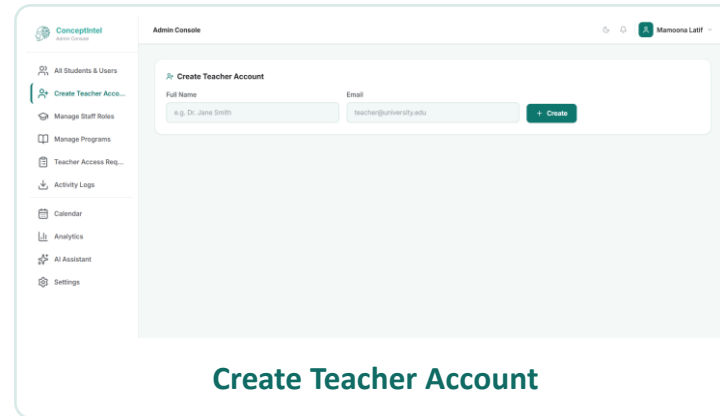
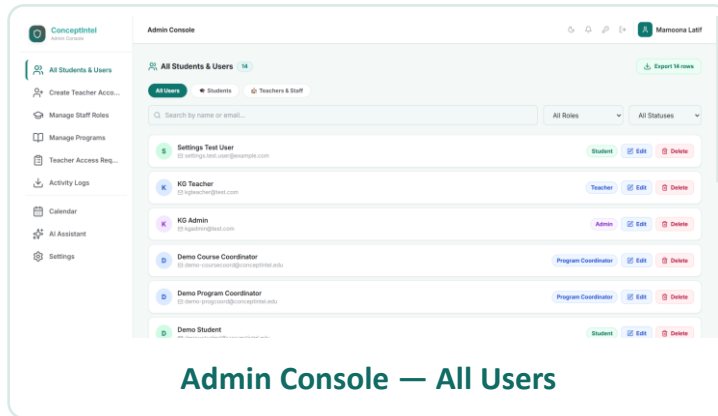
- Neo4j
- AWS S3
- Supabase Auth
- Google OAuth

Product Demonstration (1 of 2)



Live screenshots from the running prototype — every screen shown is real.

Product Demonstration (2 of 2)



Every user role — Student, Teacher, Admin — has a fully working, distinct experience.

Why ConceptIntel Is Different (1 of 2)

- AI + Knowledge Graph Fusion — concept extraction is grounded in a real graph database, never a flat AI response.
- Curriculum-Aware RAG — every answer is retrieved from the course's own uploaded material.
- Teacher-Approved AI — a dedicated Course Coordinator role reviews and approves every knowledge graph.

Why ConceptIntel Is Different (2 of 2)

- Shared Institutional Knowledge — graphs are reused across instructors teaching the same course, term after term.
- Multi-Tier Academic Governance — five distinct roles mirror how real universities actually run.
- Concept-Level Analytics — insight into exactly which prerequisite concepts a cohort is weak on, not just grades.

CURRENT STATUS

Development Progress

Completed

- Multi-Role Auth + Google Sign-In
- Course & Catalog Management
- Enrollment Module
- Content Upload + RAG Pipeline
- Knowledge Graph Engine
- Coordinator Approval Workflow
- Class Stream & Notifications
- AI Assistant Chat
- PostgreSQL Migration

In Progress

- Assignment Evaluation (grading)
- Adaptive Learning Engine (OAPA loop)

Upcoming

- Concept-Level Analytics Dashboard
- Gamification Module
- Deployment & Pilot Testing

System Limitations & Constraints

AI Extraction Isn't Perfect

Automated concept extraction can require teacher adjustment.

Prototype-Focused

Targets 3 pilot subjects, not full-scale deployment yet.

Compute at Scale

Real-time adaptation for many concurrent students needs more compute.

Depends on Material Quality

Extraction quality depends on the uploaded materials themselves.

Data Gathering Approach

Educational Materials

Slides, textbooks, and course outlines are processed with AI to extract concepts.

User Interaction Data

Quiz responses and learning progress are tracked over time.

Supervisor Guidance

Academic supervisors shape system design decisions continuously.

Existing System Analysis

Analysis of existing digital learning platforms informs every workflow.

Project Stakeholders & Roles



Project Sponsor: Air University, Islamabad

Students

Following adaptive learning paths

Teachers

Course creators and KG supervisors

Coordinators

Curriculum oversight and approval

Supervisor

Dr. Sumaira Farid — guidance

FYP Committee

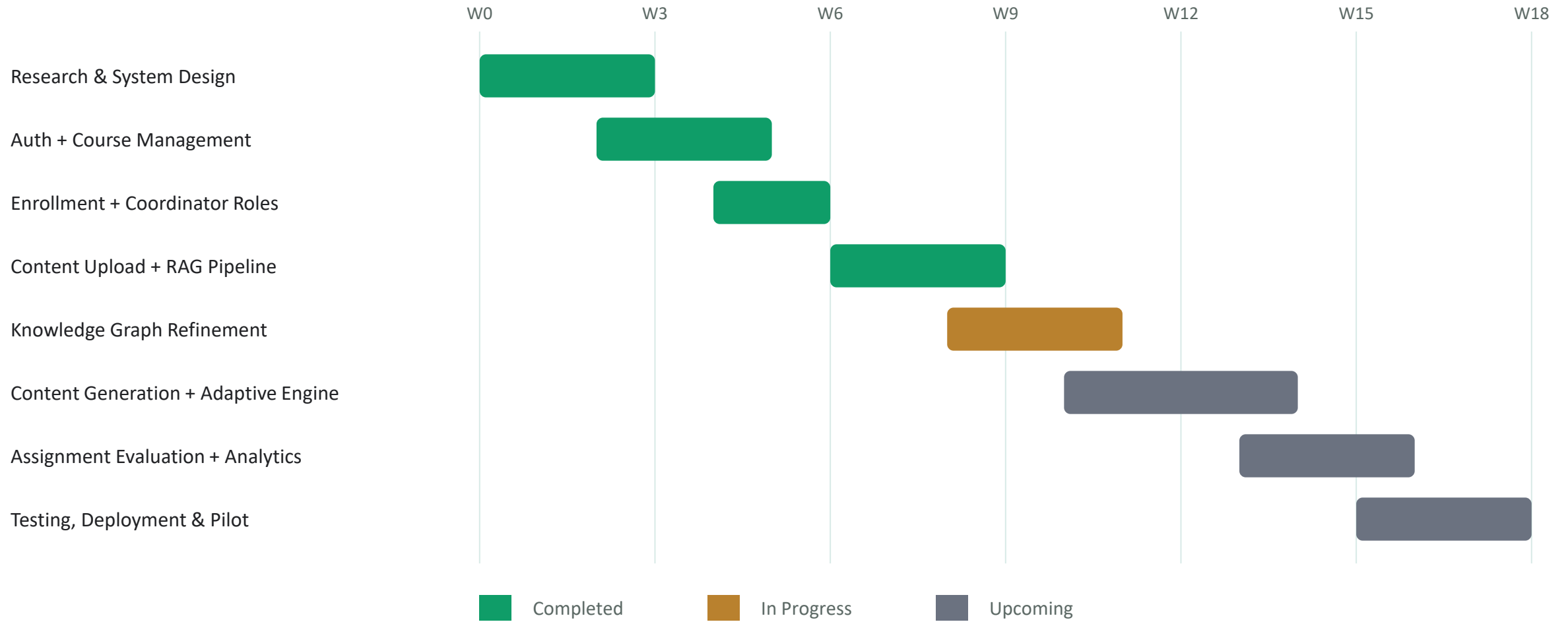
Final project assessment

Module-Based Work Division

Team Member	Assigned Modules	Key Responsibilities
Mamoona Latif Reg# 231662	User Mgmt, Content Upload, Content Gen, Notification	Auth, content uploads, AI content generation, notifications.
Laiba Reg# 231627	Course Mgmt, Knowledge Graph, Analytics, Gamification	Course management, knowledge graph, analytics, gamified learning.
Muhammad Kashif Reg# 231568	Enrollment, Supervision, Assignment Eval, Adaptive Engine	Enrollment, AI supervision workflow, evaluation, adaptive engine.

PROJECT TIMELINE

Development Gantt Chart



VISION

Impact, Scalability & Roadmap



Students

Personalized, concept-level learning



Teachers

AI-assisted teaching, full oversight



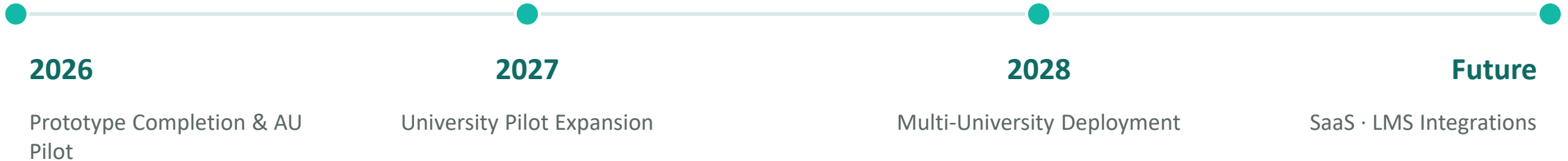
Universities

Reusable curriculum intelligence



Administrators

Data-driven academic decisions



CITATIONS

References

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2. B. P. Woolf, Building Intelligent Interactive Tutors. Morgan Kaufmann, 2009.
3. F. Doshi-Velez and B. Kim, "Towards A Rigorous Science of Interpretable Machine Learning," arXiv:1702.08608, 2017.
4. Kavitha Ganesan and L. Raj, "Educational Data Mining and Learning Analytics," arXiv, June 2017.

Conclusion

- ConceptIntel's workflow shows how users, AI modules, and course management work together seamlessly.
- Content, AI-generated guidance, and teacher supervision are fully integrated — not bolted together.
- The result: adaptive, concept-driven learning with a personalized path for every single student.

From knowledge graphs to adaptive feedback — this is intelligence built into the platform, not bolted on top of it.

THE TEAM

Meet the Team



Mamoona Latif

Reg# 231662

Auth · Content Upload · AI Content Generation ·
Notifications



Laiba

Reg# 231627

Course Mgmt · Knowledge Graph · Analytics ·
Gamification



Muhammad Kashif

Reg# 231568

Enrollment · Supervision · Assignment Evaluation ·
Adaptive Engine

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Reimagining Higher Education through Knowledge Graphs and Artificial Intelligence